

2.5.3 Operations on Data Structures

ہر ڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرنا (Traversa)...

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 21 of 24

2.5.3 Operations on Data Structures

LEARN IT

Whatever the shape, you keep meeting the same five operations:

Operation	Meaning	Quick picture
Insertion	Adding a new item into the structure.	PUSH on a stack; ENQUEUE on a queue; putting 40 into an array.
Deletion	Removing an item that is no longer needed.	POP; DEQUEUE; deleting a node from a list.
Traversal	Going through each item one by one.	Walking HEAD → ... → NULL; visiting every tree node.
Searching	Finding a specific item.	Linear search in an array; following links until 87 is found.
Sorting	Arranging items in a proper order (ascending or descending).	Bubble Sort on an array of marks.

WORKED EXAMPLE

Worked example: name the operation.

1. "Print every song from the start of the playlist." → **Traversal**.
2. "Add a print job at the rear." → **Insertion** (ENQUEUE).
3. "Undo the last typing action." → **Deletion** (POP).
4. "Is roll number 87 in the list?" → **Searching**.
5. "Arrange heights from shortest to tallest." → **Sorting**.

REMEMBER IT

Memorise **I-D-T-S-S**: Insert, Delete, Traverse, Search, Sort.

COMMON MISTAKE

Mistake: calling a PUSH "sorting" because the stack looks tidy.

Fix: PUSH is insertion. Sorting means putting values in ascending or descending order.

اردو میں سمجھیں

برڈیٹا سٹرکچر پر پانچ عمل ہوتے ہیں: داخل کرنا (Insertion)، حذف کرنا (Deletion)، ایک ایک کر کے گزرنا (Traversal)، تلاش (Searching) اور ترتیب (Sorting، صعودی یا نزولی)۔

CHECK YOURSELF

1. List the five operations.
2. Which operation is ENQUEUE?
3. Traversal vs searching — one difference?

Answers: 1. Insertion, deletion, traversal, searching, sorting. 2. Insertion (at the rear of a queue). 3. Traversal visits every item; searching stops when the target is found (or the structure ends).

Next lecture: 2.6 Identify data structures in problem scenarios